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AI Evaluation of E-commerce Chatbot Responses Using Rubric-Based Quality Scoring

Prepared by:

Patrick Adu Osei

IBM Data Analyst Professional Certificate Holder

Tools & Technologies:

Python (Pandas, Numpy) | Matplotlib/Seaborn| Jupyter Notebook | GitHub

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1. Summary

This project evaluates AI-generated e-commerce chatbot responses using a structured, rubric-based quality scoring framework. A stratified sample of 390 chatbot responses was drawn from the Bitext retail/e-commerce dataset to ensure broad category coverage. Each response was assessed across six dimensions: accuracy, relevance, completeness, consistency, business usefulness, and clarity.

The evaluation shows that chatbot performance is generally strong in terms of baseline reliability, relevance, and consistency. The dominant weakness is not factual failure, but response quality optimisation—especially overly generic phrasing, limited specificity, and occasional verbosity. These findings suggest that the chatbot is production-capable in baseline support tasks but requires refinement to improve usefulness, conversion support, and customer confidence in real e-commerce workflows.

2. Business / Analytical Question

How well do AI-generated e-commerce chatbot responses perform across key quality dimensions, and what recurring failure patterns should be prioritized for remediation to improve customer experience and business usefulness?

3. Dataset

Dataset used: Bitext retail & e-commerce chatbot dataset. The original dataset contains retail and e-commerce customer-support style prompts and AI responses with associated intent and category labels. For this evaluation, a representative stratified sample of 390 rows was selected across categories to support more reliable comparisons than the initial 10-row prototype sample.

4. Tools Used

- Python (pandas, numpy) for data handling and scoring workflow
- Matplotlib and seaborn for charts and visual outputs
- Jupyter Notebook for analysis execution and review workflow

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- CSV exports for saving scoring outputs and edge-case examples

5. Evaluation Process (What I Did)

1. Loaded and standardized the Bitext dataset by renaming key columns for evaluation clarity (instruction renamed as prompt; response renamed as ai_response).
2. Created a structured evaluation rubric using a 1–5 scale across six quality dimensions: accuracy, relevance, completeness, consistency, business usefulness, and clarity.
3. Defined an error taxonomy to capture recurring issues such as overly generic responses, missing key steps, and verbosity.
4. Built a semi-automated workflow that pre-scored responses and then prioritized manual review on low-quality rows, edge cases, and high-error clusters.
5. Aggregated scores by intent and category, computed error distributions, and generated saved visual outputs for reporting and portfolio use.

6. Key Visualizations

1. Rubric Score Distribution by Evaluation Dimension: This chart summarises average scores across the six rubric dimensions and highlights relative strengths and weaknesses in response quality.
2. Error Taxonomy Distribution: This chart shows the frequency of observed error types and reveals which quality issues dominate the evaluated sample.
3. Response Quality by Intent: This chart compares average overall response quality across chatbot intents to identify stronger and weaker use cases.
4. Response Quality by Category: This chart compares average overall response quality across business categories to highlight operational variation.
5. Edge-Case Examples Table: This output captures selected difficult prompts (for example, rude, ambiguous, or complex customer queries) together with evaluation findings and improvement suggestions.

7. Key Insights

1. The chatbot demonstrates strong baseline quality, with generally high levels of relevance and consistency across the evaluated sample.
2. The most significant weakness is overly generic phrasing. Responses are often correct and safe, but not sufficiently tailored to the user's exact request.
3. Completeness and business usefulness remain weaker than relevance and consistency, indicating that answers are often acceptable but not always maximally actionable.

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4. Verbosity appears as a secondary issue, suggesting that some responses could be made more concise without sacrificing clarity.
5. Edge-case handling appears stable in terms of tone and intent alignment, but difficult prompts still expose weaknesses in contextual adaptation and specificity.

8. Business Implications and Recommendations (Why This Project Matters)

From a business perspective, the chatbot appears reliable enough for baseline deployment in customer support scenarios. However, generic or vaguely phrased responses may reduce customer trust, limit task completion efficiency, and weaken conversion outcomes in shopping-related workflows.

- Prioritize response specificity: reduce template-like phrasing and make guidance more context-aware.
- Improve completeness in transactional flows such as payments, returns, cart actions, and order issues.
- Introduce intent-aware response tuning so that different request types receive appropriately detailed answers.
- Control verbosity by balancing clarity with conciseness, especially for common customer support interactions.
- Preserve the currently stable handling of difficult prompts while improving the usefulness of responses under edge-case conditions.

9. Challenges and How to Solve Them

- Challenge: Initial samples were too small and came from a single intent/category, making early charts and conclusions too narrow. Solution: Move to a stratified sample of 390 rows to ensure broader category coverage and more reliable comparisons.
- Challenge: Rule-based error detection was initially too strict and returned zero meaningful error labels. Solution: Relax pattern matching and use more flexible detection logic, then validate with targeted manual review.
- Challenge: Jupyter workflow interruptions caused variables such as `df_eval` and `overall_score` to disappear or be missing. Solution: Re-run the notebook in sequence using a clearly structured workflow and add validation checks before downstream analysis.
- Challenge: Some visual outputs were uninformative with tiny samples because only one intent and one category were present. Solution: Use representative sampling and save charts only after confirming the metric tables are complete and non-empty.

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- Challenge: Semi-automated scoring can misclassify nuanced responses. Solution: Prioritise human review on low-quality rows, edge cases, and rows with detected errors to improve reliability without manually scoring every record from scratch.

10. What I would Improve Next

To make the project even stronger in a evaluation setting, I would:

- Expand evaluation to additional datasets (e-commerce chatbot QA, preference comparison data, and product taxonomy data) to build a broader multi-dataset AI evaluation capstone.
- Add inter-rater agreement checks by scoring a subset twice or using a second reviewer to measure consistency.
- Refine the error taxonomy to distinguish more explicitly between genericity, omission, weak reasoning, and business-context mismatch.
- Develop a small reviewer interface or scoring form to streamline annotation and reduce manual editing overhead in notebook cells.
- Introduce benchmark comparisons between auto-scored and manually corrected results to quantify the value of human review.
- Add an optional SEO summary evaluation extension to assess AI-written KPI summaries against source metrics for quantitative readout validation.

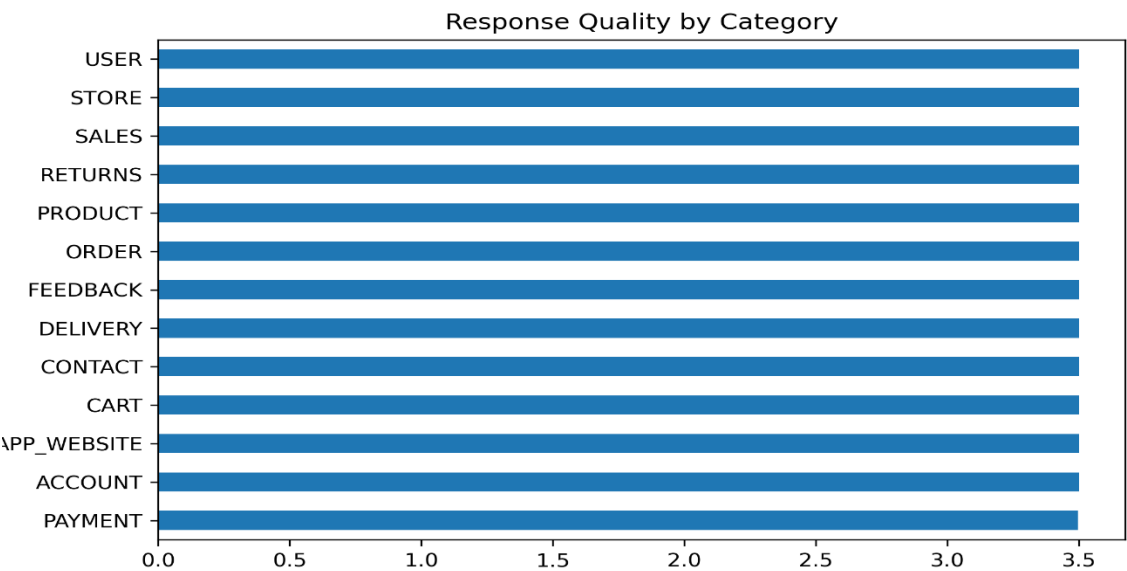
11. Files and Links

Include in repo:

- Notebook: AI Evaluation of E-commerce Chatbot Responses Using Rubric-Based Quality Scoring.ipynb
- Outputs: rubric_score_distribution.png, error_taxonomy_distribution.png, response_quality_by_intent.png, response_quality_by_category.png, edge_case_examples.csv / edge_case_examples_table.png
- Scored data: bitext_scored_results.csv (recommended final output file)
- Repository folders: /data, /notebooks, /outputs, /visuals, /report

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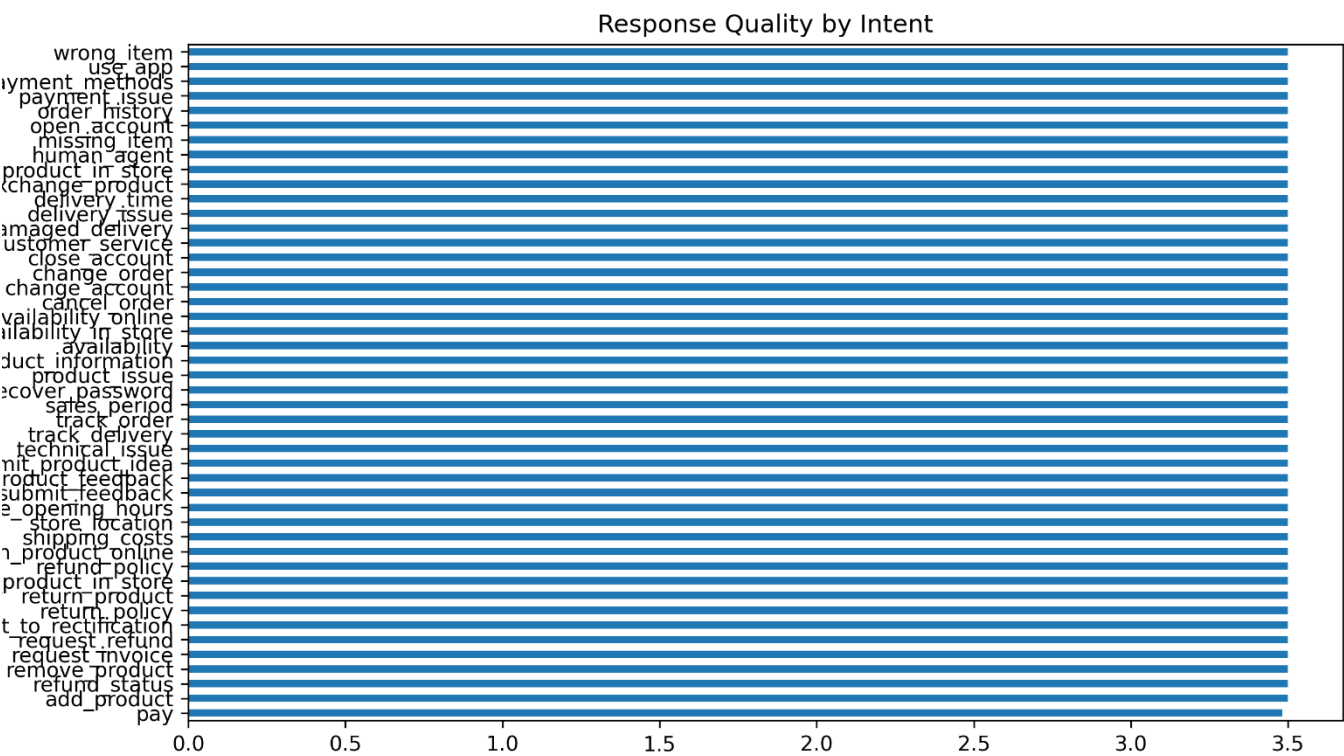
Chart 1: Response Quality by category



Insight: Category-level performance appears **stable without major breakdowns**, indicating that the chatbot maintains a broadly reliable level of quality across business areas. However, the absence of strong differentiation across categories reinforces the view that the model is consistent, but not yet highly optimized for category-specific context or business nuance.

Chart 2: Response Quality by Intent

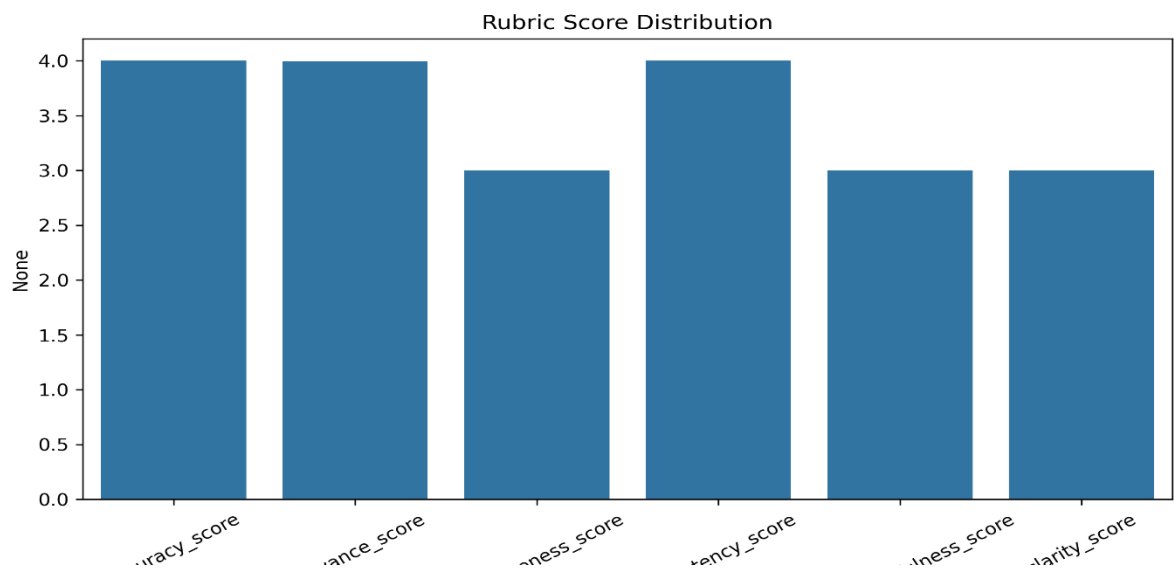
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Insight: The intent-level chart shows relatively **consistent average performance across intents**, with most scores clustering in a similar range rather than showing sharp quality differences. This indicates stable baseline reliability, but it also suggests that the model may not yet be adapting strongly enough to the specific needs of different user intents

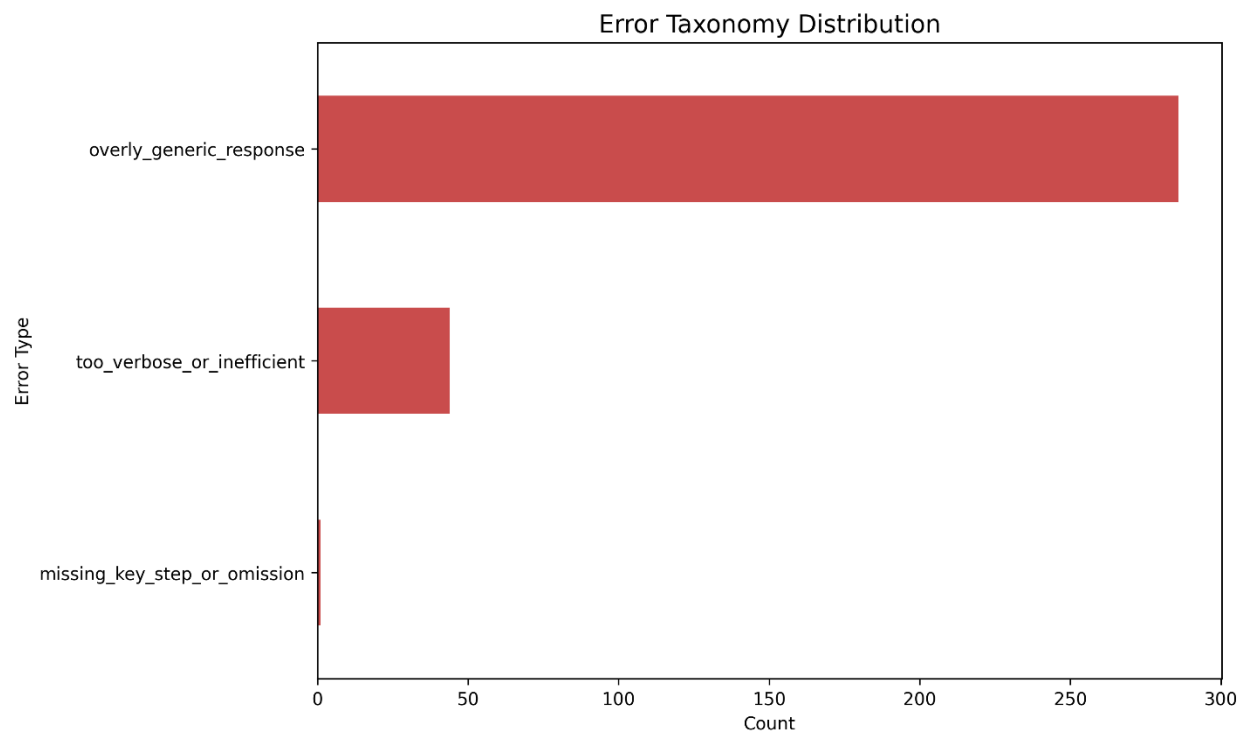
Chart 3: Rubric Score Distribution

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Insight: The rubric distribution shows that **Relevance and Consistency remain the strongest dimensions**, while **Completeness and Business Usefulness show slightly lower average scores**, indicating that although responses align with user intent, they are not always optimally detailed or actionable.

Chart 4: Error Taxonomy Distribution



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Insight: The error taxonomy visualization reveals that: (a) **overly_generic_response is the dominant error**, accounting for the majority of instances, (b) **verbosity issues** occur in a moderate number of cases, (c) **missing steps or omissions** are relatively infrequent. This shows that **the system's primary limitation lies in response quality rather than correctness or intent alignment**.

Edge-Cases

Insight: The edge-case examples show that the chatbot can remain **relevant and respectful even when prompts are rude, messy, or difficult**, which is a positive sign of behavioural stability. However, these cases also confirm that the model often defaults to **generic wording**, meaning that it handles difficult prompts safely, but not always with enough contextual depth or specificity.